

AI Voice Agent SaaS Platform

Scope of Work (SOW), Architecture & Development Plan

1. Project Overview

This project involves building a SaaS-based AI-powered voice answering system that can automatically handle customer calls, engage in natural conversations, capture user intent, and provide actionable insights to businesses.

The system will allow businesses to purchase a phone number, configure an AI agent, and automatically respond to incoming calls using real-time speech processing and AI-generated responses.

2. Objectives

- Automate customer call handling using AI
 - Capture leads, support queries, and complaints
 - Provide real-time conversation experience
 - Store transcripts and recordings
 - Deliver insights via dashboard and email
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3. Core Features

3.1 Authentication & User Management

- User Registration & Login
- Password Recovery
- Role-based access (future scope)

3.2 Subscription Management (Stripe)

- Plan-based access
- Usage limits (calls, minutes, AI usage)

3.3 Twilio Number Integration

- Buy/manage phone numbers
- Assign number to AI agent

- Handle incoming calls via webhook

3.4 AI Agent Configuration

- Company profile setup
- Define services & FAQs
- Tone selection (formal/friendly)
- Call goal (lead/support/complaint)
- AI-generated suggested conversation flow

3.5 Real-Time AI Call Handling

- Speech-to-Text (Deepgram)
- AI Processing (OpenAI)
- Text-to-Speech (Twilio TTS)
- Continuous conversation loop

3.6 Call Dashboard

- Total calls
- Call classification (Lead, Support, Complaint)
- Filters (date, type, duration)

3.7 Call Detail View

- Full transcript
- AI summary
- Extracted customer data
- Call recording playback

3.8 Notifications

- Email alerts with:
 - Transcript
 - Summary
 - Recording link

4. System Architecture

Core Components

1. Frontend (React Dashboard)
2. Backend API (Laravel)
3. Twilio Voice (Call handling)

4. Deepgram (Speech-to-Text)
 5. OpenAI (Conversation engine)
 6. TTS Engine (Twilio or external)
 7. Database (MySQL/PostgreSQL)
 8. Queue System (Redis)
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5. End-to-End System Flow

Phase 1: Call Initiation

1. Customer calls Twilio number
2. Twilio triggers webhook to backend

Phase 2: Setup

1. Identify user and load company configuration
2. Initialize call session in database

Phase 3: Conversation Loop

1. Capture user audio via Twilio
2. Convert speech to text using Deepgram
3. Send text + context to OpenAI
4. Generate AI response
5. Convert response to speech (TTS)
6. Play response back to customer
7. Repeat loop until call ends

Phase 4: Post-Call Processing

1. Store call recording
2. Generate AI summary
3. Extract key user data
4. Save transcript and metadata

Phase 5: Reporting & Notification

1. Update dashboard analytics
 2. Send email notification to business owner
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6. AI Logic Design

Inputs

- Company profile
- FAQs
- Conversation history
- User input

Outputs

- AI response
 - Intent classification
 - Data extraction (name, email, requirement)
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7. Database Structure (High-Level)

Users

- id, name, email, password

Companies

- user_id, company_name, description, tone

Numbers

- user_id, phone_number, twilio_sid

Calls

- id, user_id, type, duration, recording_url

Messages

- call_id, speaker, message

Extracted Data

- call_id, name, email, summary
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8. Development Timeline

Phase 1: Planning & Requirement Finalization (5–7 Days)

- Finalizing requirements
- Technical architecture planning

Phase 2: UI/UX Design (Figma) (10–14 Days)

- Wireframes
- High-fidelity UI design
- Design approval & revisions

Phase 3: Core Backend Development (12–16 Days)

- Auth system
- Twilio integration
- AI processing pipeline

Phase 4: Frontend Dashboard Development (10–14 Days)

- UI implementation
- Dashboard & reporting

Phase 5: AI & Voice Optimization (7–10 Days)

- Prompt tuning
- Conversation flow improvements

Phase 6: Testing & Deployment (7–10 Days)

- End-to-end testing
- Bug fixing
- Deployment

Total Estimated Time: 6–8 Weeks (~2 Months)

9. Risks & Considerations

- Real-time latency (1–2 seconds delay)
- Voice quality limitations
- API costs (Twilio, Deepgram, OpenAI)
- Scaling for high call volume

10. Future Enhancements

- CRM integrations (HubSpot, Pipedrive)
 - Multi-language support
 - Advanced analytics
 - Custom voice cloning
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11. Conclusion

This system provides a scalable and intelligent solution for businesses to automate customer communication. The architecture ensures modularity, scalability, and future extensibility while delivering a seamless AI-powered voice experience.

12. Architecture Diagram (Labeled)

Below is the high-level architecture representation of the system:

Customer → Twilio Voice → Backend API (Laravel) → Speech-to-Text (Deepgram) → AI Processing (OpenAI) → Text-to-Speech (TTS) → Twilio Response → Customer

Backend also connects to: - Database (store calls, transcripts, users) - Queue System (Redis for async jobs) - Email Service (notifications)

13. Cost Estimation (Approx)

Per Call Cost (Estimated)

- Twilio Voice: \$0.01 – \$0.03 per minute
- Deepgram STT: \$0.004 – \$0.01 per minute
- OpenAI Processing: \$0.002 – \$0.01 per interaction

Example (5-minute call):

- Twilio: \$0.05 – \$0.15
- Deepgram: \$0.02 – \$0.05
- OpenAI: \$0.02 – \$0.05

Total per call:

👉 \$0.10 – \$0.25 per call (approx)

Monthly Cost Example

If: - 1,000 calls/month - Avg 3–5 min duration

👉 Estimated monthly cost: \$100 – \$300 (infrastructure only)

Note: Does not include server hosting or scaling costs.

14. Project Milestones

Milestone 1: Planning & Setup

- Final architecture
- Project setup
- Basic authentication system

🕒 Time: 3–5 Days

Milestone 2: Core Backend & Twilio Integration

- Call handling
- Webhooks
- Basic conversation loop

🕒 Time: 10–14 Days

Milestone 3: AI Integration

- Deepgram STT
- OpenAI response system
- TTS integration

🕒 Time: 7–10 Days

Milestone 4: Dashboard & UI

- Call logs
- Filters
- Call detail pages

 Time: 7–10 Days

Milestone 5: Testing & Deployment

- Bug fixing
- Optimization
- Production deployment

 Time: 5–7 Days

15. Final Notes

- Recommended to start with MVP and iterate
 - Real-time voice systems require optimization post-launch
 - Cost scales with usage (SaaS-friendly model)
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Prepared By: AI Solutions Architect